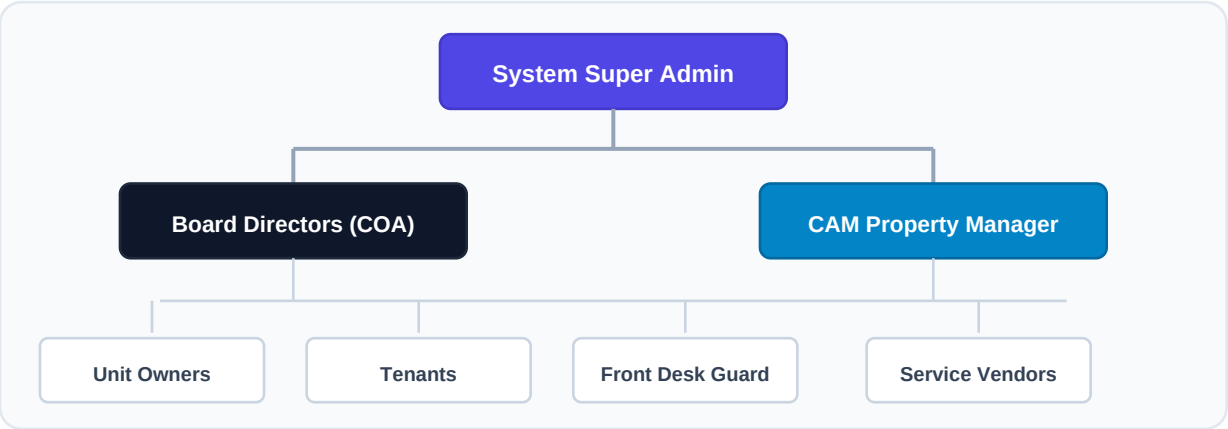


High Level Requirement & Architecture Document for Condo Management (COA)

Project: Condominium Owners Association (COA) Platform | Client: Produtions LLC | Marketplace: US & Global

System Architecture & User Role Mapping Overview:



Section 1: Data Security, Privacy & PCI Compliance

- **Email & Phone Verification:** Email address must be verified. Phone number verification via SMS/OTP is mandatory before adding bank accounts or credit cards.
- **PCI-DSS Vault Encryption:** Bank account and credit card details must be tokenized and encrypted, displaying only the last 4 digits.
- **Payment Transaction Security:** Every payment transaction requires SMS/OTP step verification for authorization.
- **Data Isolation (Multi-Tenancy):** COA Members and Board Directors cannot see data of other COAs. Admin cannot view unassociated communities.
- **Audit Trail:** System must maintain encrypted audit logs for logins, role changes, and financial transactions.

Section 2: High-Rise Condo Operations & Scope

- **US Condominium Act Compliance:** Meets requirements for Florida Chapter 718, Davis-Stirling Act, and Fair Housing Act (FHA).
- **Shared Asset Management:** Specialized tracking for elevators, reserve funds, parking bays, storage lockers, and package rooms.
- **Multi-Building Support:** Support for multi-tower complexes sharing central clubhouse and gate facilities.

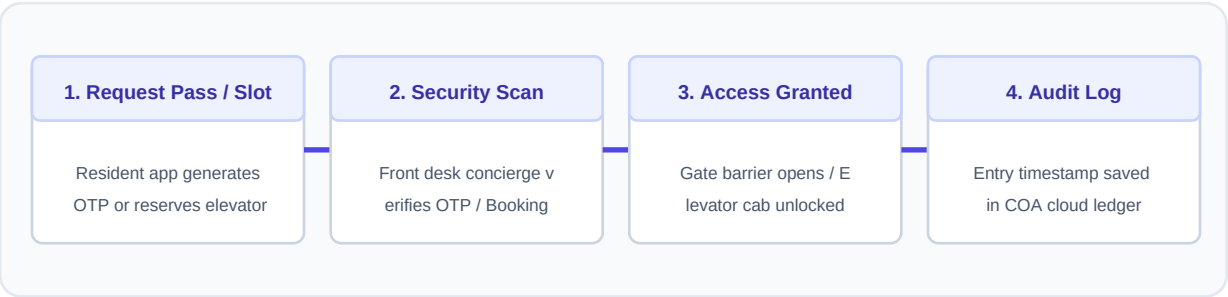
Section 3: Types of Condo Clients & Web Experiences

- **Central Portal vs Whitelabel:** Clients can use central portal (vhoas.com/condo) or dedicated custom whitelabel domains.
- **Verification Access Tiers:** Non-verified members have restricted view; verified unit owners/tenants have full feature access based on assigned profile.
- **Role Switching:** Multi-unit owners or board members can seamlessly toggle between Board View and Resident View.

Section 4: User Profile & Role Permissions Matrix

Role	Target User	Key Permissions & Capabilities
Super Admin	VHOAS Team	Global platform configuration, client onboarding, billing, system audit logs.
CAM Property Manager	Licensed Agency	Work orders, Estoppel Certificates, vendor dispatch, dues invoicing, COI verification.
Board Directors	President/Treasurer	Operating budget approval, reserve funds, E-voting, board meeting minutes, bylaws.
Unit Owner	Homeowner	Monthly dues payment, board proxy voting, ARC remodel requests, tenant invitations.
Resident / Tenant	Occupant	Visitor OTP passes, freight elevator booking, package tracking, amenity bookings.
Front Desk Concierge	Lobby Security	Guest OTP scanner, delivery parcel logging, move-in elevator verification, guest parking tags.
Service Vendor	Contractors	Work order status updates, invoice submission, COI document upload.

Visitor & Elevator Reservation Workflow:



Section 5: High-Rise Gate Access & Visitor OTP System

- **Guest Pass Generation:** Residents generate 6-digit OTPs or QR passes for guests and delivery drivers via app.
- **Concierge Verification Interface:** Front desk guard scans or enters OTP to validate visitor name, unit destination, and vehicle tag.
- **Real-time Entry Alerts:** Resident receives instant app push notification when guest enters lobby.

Section 6: Freight Elevator & Move-In / Move-Out Reservation

- **Slot Reservations:** Unit owners/tenants book 2-hour or 4-hour freight elevator slots prior to moving truck arrival.
- **Move Deposit Escrow:** Automated \$250 refundable damage deposit hold via credit card prior to slot confirmation.
- **Pre/Post Checklist:** Property manager completes digital inspection log before returning deposit.

Section 7: Front-Desk Parcel Room & Delivery Logging

- **Barcode Delivery Scanning:** Concierge scans package tracking barcodes (FedEx, UPS, Amazon) auto-matching unit resident.
- **Automated SMS/Email Alert:** Immediate notification sent to resident with pickup QR code.
- **Digital Sign-Out:** Resident signs touchscreen or scans QR at front desk upon package collection.

Section 8: Financial Ledger, Dues Collection & Reserve Funds

- **US ACH & India Bank Flows:** Supports US ACH auto-debit via Stripe/Plaid to COA Bank Account, and Indian local bank flows for global properties.
- **Operating vs Reserve Fund Accounting:** Automated split of incoming assessment dues between General Operating Fund and Capital Reserve Account.

- **Automated Late Fees:** Grace period logic (e.g. \$50 fee after 5th of month) auto-added to checkout balance.

Section 9: Estoppel Certificates & Resale Disclosures

- **Title Company Portal:** Real estate title companies request official Estoppel certificates online.
- **Instant Generation:** Auto-populates outstanding dues, pending special assessments, and open violations for title clearance.

Section 10: Violation Tracking & Fine Appeal Hearings

- **Violation Form:** Photo attachments, date, description, and fine amount generated by Board or CAM Manager.
- **Status Tracking:** Open, Disputed, Paid, Cancelled.
- **30-Day Dispute Window:** Member can submit dispute appeal with evidence within 30 days for Board hearing.

Section 11: Technical Specifications & AWS Infrastructure

- **Frontend:** React, Vite, Tailwind CSS, Axios.
- **Backend:** Python FastAPI, SQLAlchemy ORM, PostgreSQL (AWS RDS).
- **Security:** OAuth2 JWT tokens, Captcha integration on auth pages, SSL encryption, PCI-compliant payment gateways (PayPal/Stripe).
- **Modular Codebase:** Tenant-isolated schemas, reusable UI components, whitelabel branding support.